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The Agent Control Plane

Why Distributed AI Agents Need Telecom's Two-Plane Architecture, SIP for Control, QUIC for Carriage

Why specialized agents, geographically dispersed across providers and jurisdictions, are a **network** and not a function call, and why the proven separation of a **signaling plane** from a **delivery plane** is the architecture they are missing.

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Executive Summary

Artificial intelligence is no longer a single model behind a single endpoint. It is becoming a **distributed system of specialized agents**, reasoning here, retrieval there, a vision model in one region, a code model in another, a procurement agent calling a pricing agent calling a vendor's agent. These agents are geographically dispersed for latency, data residency, capability, and cost. They span multiple providers and multiple jurisdictions. They are increasingly mobile and ephemeral, spun up at the edge and torn down moments later.

Yet the industry is building this distributed system on the thinnest possible substrate. Stateless HTTP request and response. Remote procedure calls. That abstraction quietly assumes the wrong thing, that an agent interaction is a function call with a return value, when in practice it is a **session**. Stateful, long-lived, multi-party, mobile, and federated across administrative domains that trust each other only conditionally.

Telecommunications confronted exactly this topology decades ago and answered it with one foundational architectural decision. **Separate the control plane from the data plane**. Signaling establishes, routes, authenticates, negotiates, modifies, and tears down a session. A separate bearer carries the payload. That split is why a call can be addressed, located, authorized, metered, secured, and handed between carriers independently of the bytes it carries.

This paper argues that distributed AI agents need the same two-plane separation, and that two battle-hardened protocols already provide it: **SIP for the control plane and QUIC for the delivery plane**. To make the inheritance concrete, it sketches five primitives, four minted from proven telecom ancestors, and one that has no ancestor at all, marking the precise point where the agent world departs from the telephone network:

- **Agent Session Protocol (ASP)**, what we call the control plane for agents: establishing, routing, negotiating, and terminating agent-to-agent sessions. *Heir to SIP*.
- **Agent Location Service (ALS)**, the registry that answers “which agent, with which capabilities, reachable where, right now.” *Heir to the SIP registrar and the telecom location register*.
- **Agent Border Controller (ABC)**, the policy, security, and topology demarcation between two agent domains. *Heir to the Session Border Controller*.
- **Agent Bearer**, the encrypted, multiplexed, migration-tolerant delivery plane that carries the actual payload. *Heir to the media/bearer plane, realized on QUIC*.

- **Signaling Moderator (SigMod)**, the element that listens on the bearer for session-control intent expressed *inside the content* and drives the signaling plane from it. *It has no telecom ancestor, and that absence is the central finding of this paper.*

This is the companion to *The Enterprise AI Accountability Framework*. That paper built the **accounting plane**, the Token Detail Record, Proxy of Record, Model Routing Engine, and Model Decision Record. This paper builds the **control and delivery planes those records ride on**. The connection is not incidental. In telecom, the billing record was always born from a signaling event. Accountability is a child of the control plane.

The Agent Topology Is a Network, Not a Function Call

The mental model most teams carry into agent design is the API call: send a request, receive a response, done. It is a comfortable model because it is the one the web was built on, and for a single prompt to a single model it is adequate. It stops being adequate the moment intelligence is decomposed into specialists that must cooperate.

A mature agentic workflow does not look like a function call. It looks like a **conversation among distributed parties**. A planning agent engages a retrieval agent, which consults a domain agent in another region, which calls a tool exposed by a third party's agent under a different security policy and a different legal regime. The work is long-lived, it carries state, it fans out and rejoins, and it crosses ownership boundaries. Participants appear and disappear. Some run at the edge for a few hundred milliseconds and are gone. The interaction has a beginning, a middle, and an end, it is a session, not a request.

Distributed systems of this shape have well-known, unglamorous problems that request/response simply does not address: **how do you name a party, find where it currently is, route to it across a domain you do not control, negotiate what it can do, authenticate it, keep the session alive when it moves networks, and tear it down cleanly?** These are not AI problems. They are session problems, and they were solved at planetary scale by an industry that has been doing nothing else for forty years.

What Telecom Already Solved: Separating Signaling from Bearer

The telephone network learned early that the hard part of a call is not carrying the voice, it is everything around the voice. Setting it up, finding the called party, choosing a route, authorizing

it, negotiating what both ends support, holding it together as conditions change, billing it, and tearing it down. So the network was split into two planes that evolve independently.

The **control plane**, signaling, carries the intelligence of the session. It establishes who is calling whom, locates the destination, selects a path, negotiates capabilities, enforces policy, and signals every state change until teardown. First SS7 in the circuit-switched world, then the Session Initiation Protocol (SIP) in the IP world, became the lingua franca of this plane. The **data plane**, the bearer, carries the actual payload, classically as RTP media streams, and does exactly one job. Move the bytes, fast.

The power of the separation is that the two planes can be reasoned about, secured, scaled, and operated on their own terms. You can route and authorize a session without touching its content. You can meter it from signaling events alone. You can hand it from one carrier to another at a well-defined border. You can hold the session identity stable while the underlying transport changes beneath it. None of this is possible when control and data are fused into one stateless exchange, which is precisely the condition most agent architectures are in today.

Address the party, establish the session, negotiate capability, carry the payload, secure the border.

That line is the spine of this paper. Each clause is a problem distributed agents already have and have not yet named, and each maps directly onto a telecom mechanism that has been hardened in carrier production for decades.

Why SIP

The control plane, already written.

SIP is not a telephony protocol that happens to be reusable. It is a **general session-control protocol** that was proven first in telephony. Strip away the voice associations and what remains is an almost uncanny match for the unsolved problems of multi-agent systems. Consider what SIP already standardizes:

- **Identity and addressing.** The SIP URI (*agent:role@domain*, in spirit) gives every endpoint a routable, federated, human-legible address that is independent of its current network location, exactly the naming layer agents lack.
- **Location and registration.** A registrar tracks where a given identity is reachable *right now*. Ephemeral and mobile agents need precisely this: a live answer to “where is this capability at this instant,” not a hard-coded URL that was true at deploy time.
- **Routing through intermediaries.** Proxies, redirect servers, and back-to-back user agents route a session across domains under policy, without the endpoints needing a direct, pre-arranged path. This is the agent-federation problem already solved.
- **Capability negotiation.** The offer/answer model (SDP, in SIP’s case) lets two parties agree on what they mutually support before committing, codecs in telephony. In the agent world, models, tool schemas, context limits, modalities, and protocol versions.
- **Authentication and federation across domains.** SIP grew up in the carrier-interconnect world, where independently operated networks with their own policies must trust each other conditionally and verifiably. That is the multi-tenant, multi-vendor agent world, described exactly.
- **Forking, presence, and session modification.** Ring several candidates and take the first to answer. Know whether a party is available before engaging. Renegotiate mid-session as needs change. Each has an obvious agent analogue, racing redundant agents, capability discovery, and live re-routing.
- **Observability and lawful interception.** Because signaling is separate and explicit, sessions are inspectable, auditable, and, where legally required, interceptable, by design rather than by bolt-on. In a regulated, multi-jurisdiction agent economy this is not optional, and it is already mature here.

The point is not that agents should literally place phone calls. It is that the **session-control semantics agents need have already been specified, interoperably implemented, and carrier-hardened** under the name SIP. Reinventing them as proprietary JSON over HTTP is not innovation. It is rediscovery, slower and less complete.

Why QUIC

The bearer, modernized.

If SIP answers *how the session is controlled*, QUIC answers *how the payload is carried*. A control plane is only as good as the delivery plane beneath it, and the delivery characteristics of distributed agents are demanding in ways classic HTTP/TCP serves badly. QUIC, the transport beneath HTTP/3, was built for exactly this profile:

- **Encryption is mandatory, not optional.** QUIC integrates TLS 1.3 into the transport itself. Every agent session is confidential and integrity-protected by construction, the baseline a multi-jurisdiction, multi-tenant economy must have and cannot retrofit cheaply.
- **Stream multiplexing without head-of-line blocking.** A single agent exchange is rarely one stream. It is many concurrent ones, token deltas, tool calls, retrievals, control messages. QUIC carries them as independent streams over one connection, so a stall in one does not freeze the rest. TCP cannot make that promise.
- **Connection migration.** A QUIC connection survives a change of network path, the client's address can change and the session continues. For agents that are mobile, edge-resident, or shifted across regions for residency or failover, this is session continuity under mobility, delivered at the transport layer.
- **Low-latency establishment.** 0-RTT and 1-RTT handshakes cut setup cost, which matters enormously when a single user action fans out into hundreds of short-lived agent sessions rather than one long call.
- **Per-stream flow control and modern loss recovery.** Fine-grained backpressure and better congestion behavior suit the bursty, parallel, geographically stretched traffic that agent meshes generate.

QUIC is, in effect, the bearer plane rebuilt for an encrypted, mobile, massively multiplexed world, the modern heir to RTP's job, with the security and mobility the original media plane never had. And the convergence is already visible. The standards community is defining SIP *over* QUIC, which is the two planes of this paper meeting in a single modern stack. Telecom's architecture is not being left behind by the new transport. It is being re-expressed on top of it.

The Primitives

Four telecom mechanisms re-minted for agents, and one with no ancestor.

As with the accountability framework, naming the primitives is what lets an industry build them deliberately rather than rediscover them by accident. Four carry the control and delivery planes, inherited intact from telecom. The fifth, treated in its own section below, is the single deliberate departure.

ASP The **Agent Session Protocol** is what we call the control plane for agents, the SIP-descended layer that addresses, establishes, routes, negotiates, modifies, and tears down a session between agents, independent of the payload and independent of the agents' current locations.

ALS The **Agent Location Service** is the live registry that resolves a stable agent identity to its current reachable endpoint and advertised capabilities, heir to the SIP registrar and the network location register. It is what makes ephemeral, mobile agents addressable without hard-coding where they happen to be.

ABC The **Agent Border Controller** is the demarcation point between two agent domains. It enforces policy, hides internal topology, terminates and re-originates sessions, and applies security at the boundary, heir to the Session Border Controller that polices every carrier interconnect. It is the natural twin of the accountability framework's Proxy of Record. The same border, viewed from the signaling-and-security side rather than the accounting side.

BEARER The **Agent Bearer** is the delivery plane itself, the encrypted, multiplexed, migration-tolerant carriage of payload, realized on QUIC. It does one job supremely well: move the bytes, securely and without head-of-line stalls, even as endpoints move.

SIGMOD The **Signaling Moderator** is the one primitive with no telecom ancestor. It is the listener on the bearer that detects session-control *intent* expressed inside the payload, "task complete," "hand this to the billing agent," "stop" and translates it into a signaling action on the control plane. It is, by deliberate design, the **single point** at which the two planes touch. The next section explains why this primitive must exist for agents and could not exist in telephony.

The Primitive With No Ancestor: When Intent Lives in the Content

Where the agent world departs from the telephone network, the one place the analogy must break.

Every primitive so far has been an inheritance. This one is not, and the reason it is not is the most important architectural observation in this paper.

In telecommunications, the intent to control a session never travels inside the session. When a person ends a call, they do not say the word “goodbye” and have the network act on it, they press a button, replace a handset, send an explicit, out-of-band on-hook signal. The bearer carries voice and only voice. It has no authority over the session’s lifecycle. You can speak any sentence into a live call, “hang up,” “transfer me,” “end the call” and nothing happens, because the content plane was deliberately built deaf to its own meaning. Control intent originates with a human acting *on* the signaling plane directly, never *through* the bearer. That separation is not an accident of telephony. It is the very thing that makes the two-plane model clean.

Agents break this assumption at the root. An agent’s intent to end a session, fork a new leg, hand the task to another specialist, or escalate to a human is **expressed in the content itself**, in the natural-language payload flowing across the bearer. “The analysis is finished, route the result to the billing agent” is simultaneously the message *and* a teardown-plus-transfer instruction. Unlike the telephone caller, the agent has no button to press and no handset to hang up. Its only channel of expression is the bearer. The intent to control the session is born on the data plane and nowhere else.

This is the precise point where a strict telecom architecture, applied naively, would fail. If the control plane is rigidly separated from the bearer, as telecom demands, and as this paper otherwise insists, then no one is listening where the intent actually lives, and the session can never be driven by the work it is doing. Something must read the content plane for control meaning and act on the signaling plane in response. That something is the **Signaling Moderator**.

A person cannot say “goodbye” and have the call drop. An agent can, and that single fact is why one new primitive is needed.

SigMod is a bearer-plane listener with a signaling-plane mandate. It observes the payload, extracts the embedded control intent, completion, handoff, fork, escalation, termination, and issues the corresponding ASP action. Modify the session, originate a new leg through the Agent Border Controller, or tear down cleanly. It is the deliberate, governed re-coupling of the two planes that telecom spent forty years keeping apart.

And precisely because that separation is so valuable, the re-coupling must be surgical. SigMod is the **single, exclusive point of interaction between the two planes, and there is no other**. Everywhere else, the discipline holds without compromise. Signaling controls, the bearer carries, and the two do not touch. The planes are re-joined at exactly one designated, auditable element and nowhere else. This is the minimum possible divergence from the proven model, one

primitive, one interaction point, introduced only because the nature of agent intent leaves no alternative. Make the coupling promiscuous, let any component anywhere act on content it happens to read, and the entire value of plane separation evaporates. Sessions become un-auditable, control becomes non-deterministic, and the architecture collapses back into the stateless tangle it was meant to replace.

Concentrating the coupling in one named element is what keeps it governable. Because SigMod is the only bridge, it is the only place that must be secured, audited, and constrained. It is the one choke point where “the content asked for an action” becomes “the session took an action.” It is where prompt-injection defenses, authorization checks, and policy on which intents may be honored from which parties all naturally live. One door, watched closely, rather than a thousand windows left open. That SigMod has no telecom ancestor is not a weakness in the analogy, it is the analogy correctly identifying its own single exception, and handling it with the same discipline telecom would.

Why Overlay a Telecom Delivery Plane on Dispersed Agents

The objection writes itself. Agents are software, the web already connects software, why import telecom machinery? The answer is that the *topology* of specialized agents is a telecom topology, and topology dictates architecture.

Agents are dispersed for the same reasons networks are: **latency** (run the model near the user or the data), **sovereignty and residency** (keep inference and data inside a jurisdiction), **capability locality** (the right specialist lives where it lives), and **cost** (place work where it is cheapest to perform). Dispersion across independently operated domains is not an accident to be engineered away. It is the steady state. And a steady state of cooperating, independently operated, policy-bearing domains is the definition of an **interconnect**, the problem telecom has spent its entire existence solving.

Once that is admitted, the inheritances follow directly. Capability negotiation across vendors: solved by offer/answer. Identity and trust across administrative domains: solved by federated signaling and interconnect security. Session continuity as parties move or fail over: solved by connection migration and re-routing. Topology hiding and policy enforcement at the boundary: solved by the border controller. Auditability and lawful observation across jurisdictions: solved by an explicit, separable control plane. Each of these is a problem multi-agent systems are walking into right now, one production incident at a time, and each already has a hardened answer that predates the problem.

Building the agent mesh on bare HTTP does not avoid these problems. It merely defers them, and guarantees that each team solves them again, privately, incompatibly, and later, the precise failure mode the accountability paper warned about for records, now repeated for sessions.

What About MCP and A2A?

They are the application layer, and they ride on the planes this paper describes.

Any architecture proposed for agent interconnect now has to answer for two protocols that already exist, ship in production, and are backed by every major cloud. They are the **Model Context Protocol (MCP)** and **Agent2Agent (A2A)**. They are not strawmen, and the honest position is not to dismiss them, it is to locate them. They solve real problems, and they solve them at a different layer than this paper does.

The two divide cleanly. **MCP is vertical**. It standardizes how a single agent reaches the tools, data, and resources it needs, the integration layer between a model and its capabilities. It has effectively won that layer. **A2A is horizontal**. It lets two agents that do not share a codebase discover each other through signed *Agent Cards* and delegate tasks across organizational lines. Both are application-layer protocols, and both ride on the same substrate, stateless HTTP with JSON-RPC and server-sent events. They standardize **discovery and invocation**. They name what an agent can do, and hand it a task.

What they do not standardize is everything beneath the task. **A task is not a session**. It has no lifecycle that survives a change of network, no native fork-and-rejoin, no in-flight renegotiation, no clean teardown semantics independent of the transport carrying it. **An Agent Card is a directory entry, not a live location register**, it answers “what can this agent do and where was it published,” not “where is this ephemeral capability reachable at this instant,” which is the ALS problem. **HTTP is not a bearer** built for encrypted, head-of-line-free, migration-tolerant carriage. That is the QUIC problem. And A2A’s own task handoffs **carry control intent inside the task payload**, which is precisely the SigMod problem, inherited and not yet named.

MCP and A2A decide *what* to do and *to whom*. The control plane governs how the session lives. The bearer governs how it travels. Those are different layers, and the agent stack needs all of them.

So the relationship is composition, not competition. A2A discovers a partner and delegates a task. The **Agent Session Protocol** establishes and controls the session that delegation runs inside. **QUIC** carries it. The **Agent Border Controller** polices the boundary between

domains. The **Proxy of Record** mints the record. MCP and A2A sit *on top of* the two planes, not against them. They are the layer that frames the call. The planes are the layers that carry, secure, move, and account for it.

There is a deeper signal here, and it favors the argument rather than undercutting it. A2A's evolution toward **signed Agent Cards and verifiable cross-domain trust** is the discovery layer rediscovering, bottom-up and one release at a time, exactly what federated signaling systematized decades ago. Identity, registration, and conditional trust between independently operated domains. That is this paper's thesis playing out in real time. The application layer is converging, by necessity, on the problems the control plane already solved, and it will arrive there faster by inheriting the answers than by deriving them again.

How the Two Planes Compose with Accountability

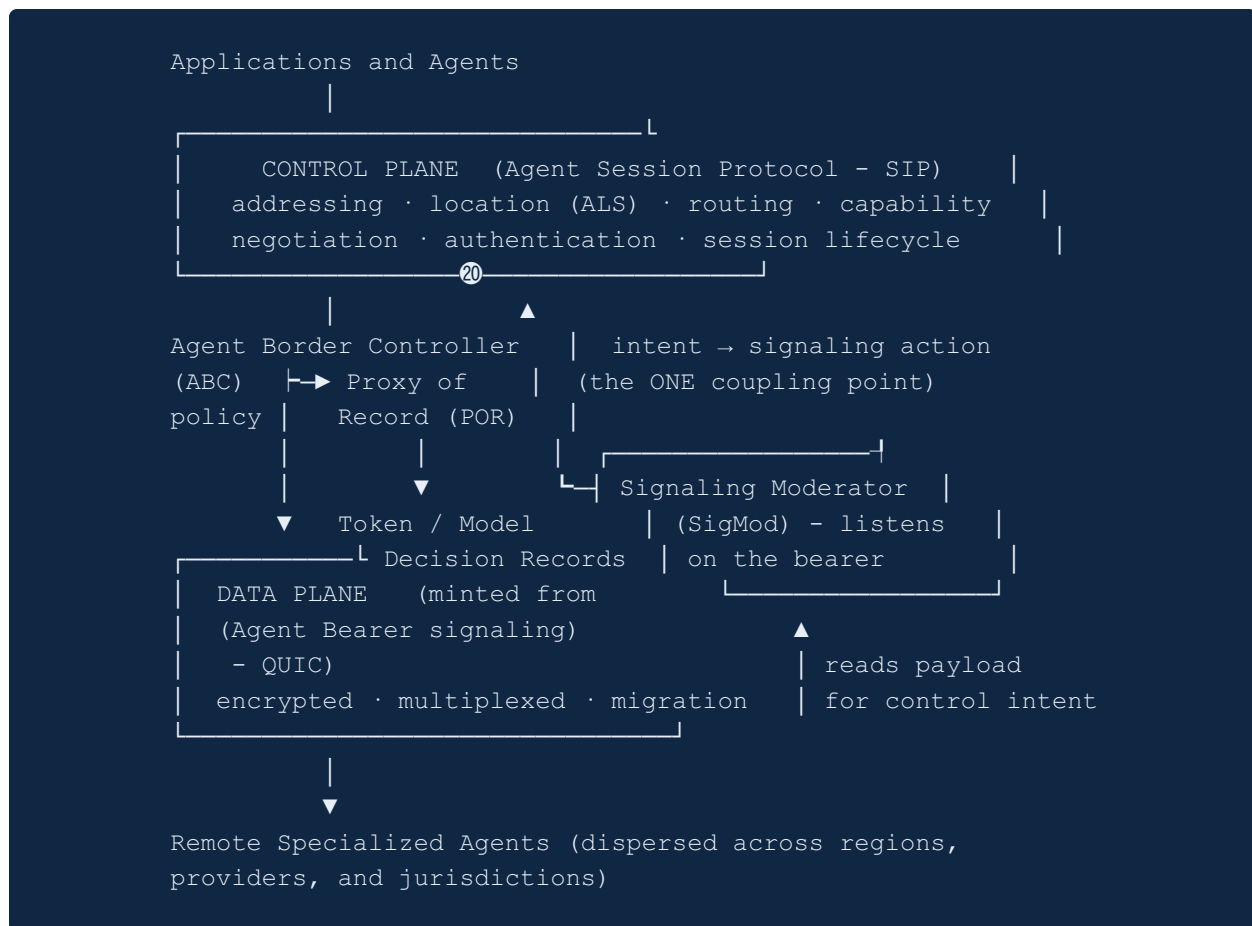
The strongest argument for this architecture is that it closes the loop with the accountability framework rather than sitting beside it. In telecommunications, the Call Detail Record was not produced by the bearer, it was produced by **signaling**. Call setup, answer, and teardown are signaling events, and the billing record was minted from them. The control plane is where accountability is born.

The same will hold for agents. The **Model Routing Engine** is a control-plane function. It selects a destination under policy, which is a routing decision in the ASP. The **Model Decision Record** is the provenance of that signaling decision, why this route, this agent, this model. The **Token Detail Record** is minted at session events observed at the border, and that border is exactly where the **Agent Border Controller** and the **Proxy of Record** coincide. One element, two faces. Signaling and security on one side, recording and reconciliation on the other.

This is why the two papers are one program. Accountability without a control plane is records with no native source. A control plane without accountability is sessions that no one can bill, audit, or reconcile. Telecom built them together, in that order, signaling first, records minted from it, and the agent economy will, knowingly or not, do the same.

Reference Architecture

The two planes compose, with the accounting plane drawing from signaling events at the border, and with SigMod as the one, and only, place the bearer feeds the control plane:



The control plane decides and signals. The data plane carries. The border is where both security and accountability are applied. The shape is telecom’s, with a single deliberate addition. **Sig-Mod** taps the bearer, reads the embedded control intent, and drives the signaling plane, the one and only point where the two planes touch. Everywhere else the separation is absolute. Only the payload has become inference instead of voice.

Conclusion

The agent era is being built as if every interaction were an isolated request, when the reality on the ground is a **federated network of stateful, mobile, multi-party sessions** crossing domains that trust each other only as much as they can verify. That is not a new kind of system. It is the oldest kind of distributed system there is, and the discipline for operating it was written, debugged, and hardened over forty years of carrier production.

The lesson is the same one the accountability framework drew, applied to a different plane. **Do not reinvent what telecom already proved.** Separate the control plane from the data plane. Use SIP’s session-control semantics for addressing, location, routing, capability negoti-

ation, federation, and lifecycle. Use QUIC's encrypted, multiplexed, migration-tolerant transport as the bearer. Mint the accountability records from the signaling events, at the border, exactly where telecom always minted them.

Inherit everything that transfers, and be honest about the one thing that does not. Agent intent, unlike a caller's, lives inside the content, so the planes must be re-joined at exactly one governed point. The **Signaling Moderator**, the bearer-plane listener that turns intent into signaling. One primitive, one coupling point, no other, the minimum possible divergence from a proven model, introduced only because the nature of agents demands it. That single, disciplined exception is what lets every other part of the architecture stay exactly as telecom left it.

SIP and QUIC are not legacy baggage to be left behind by the AI era. They are the most thoroughly tested answers in existence to the exact problems multi-agent systems are about to hit at scale, identity, location, federation, mobility, security, and observability across independently operated domains. The enterprises that operate their agents as a network, with a real control plane and a real delivery plane, will run a system they can route, secure, audit, and reconcile. The ones that keep firing stateless requests into the dark will spend the next several years rediscovering, painfully and privately, what a dial tone already knew.

Intelligence will be distributed.

Distributed intelligence is a network, so operate it like one.

About the author. **Jerry W. Lambert, Sr.** brings forty years of hyper-volume operations experience spanning billing, settlement, routing, and business intelligence across platforms that process extremely large data volumes, including thirty years at the executive level for private and public companies. He holds two U.S. patents. Contact: me@jerrywlambert.com | <https://jerrywlambert.com>